2. Employee Roles, Competency Matrix & Site Safety Manual

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Primary OWL Classes: `Employee`, `SiteManager`, `Designer`, `Architect`, `ConstructionWorker`, `Electrician`, `Plumber`, `ManagementRole`, `ManualWorker`
Primary OWL Properties: `worksOn`, `hasProject`, `hasEmployee`, `belongsToProject`, `usesMaterial`

2.1 Taxonomy of Employees in the Construction Ontology

Every staff member at ApexBuild Construction Corp. is an instance of `Employee`, which is a direct subclass of `Person`. In the OWL ontology, `Employee` inherits `worksOn some Project` from `Person` and adds the restriction `hasProject some Project`. Employees are divided into supervisory/design specialists and on-site manual execution staff:

- **`SiteManager` (Stavbyvedúci):** Subclass of `Employee` and member of `ManagementRole`. Must satisfy:
  - `hasEmployee some Employee` (a Site Manager supervises subordinate employees on site).
  - `hasProject only Project` (`owl:allValuesFrom Project` – all projects managed by a Site Manager must be valid instances of `Project`).
  - `worksOn some Project` (actively works on assigned construction projects).
- **`Architect` (Architekt):** Subclass of `Employee` and member of `ManagementRole`. Must satisfy:
  - `hasProject some CommercialProject` (every Architect in our company is assigned to at least one commercial project, such as an `OfficeBuilding`).
  - `hasCustomer some Customer` (directly consults with the project client).
  - `worksOn some Project`.
- **`Designer` (Projektant):** Subclass of `Employee`. Specializes in residential engineering and interior/structural layouts. Must satisfy:
  - `hasProject some ResidentialProject` (every Designer is responsible for at least one residential project, such as a `FamilyHouse` or `ApartmentBuilding`).
  - `worksOn some Project`.
- **`ConstructionWorker` (Stavebný robotník):** Subclass of `Employee` representing field personnel who execute physical works on site. Must satisfy:
  - `belongsToProject some Project` and `worksOn some Project`.

2.2 Specialized Manual Workers (Electricians and Plumbers)

Within the class `ConstructionWorker`, the ontology defines two certified technical trades that together form the equivalent union class `ManualWorker`:

> **Formal OWL Axiom (Equivalent Class):**
> `ManualWorker ≡ Electrician ⊔ Plumber` (The union of `Electrician` and `Plumber`).

- **`Electrician` (Elektrikár):** Subclass of `ConstructionWorker`. Defined with three key existential restrictions:

1. `worksOn some BuildingProject` (Electricians work on broad building projects, including residential, commercial, and industrial buildings).
2. `usesMaterial some Material` (handles electrical cabling, steel conduits, insulation materials).
3. `worksOn some Project`.

- **`Plumber` (Inštalatér / Vodár):** Subclass of `ConstructionWorker`. Defined with three existential restrictions:
  1. `worksOn some ResidentialProject` (Plumbers at ApexBuild are specifically mandated to work on residential projects such as `FamilyHouse` and `ApartmentBuilding`).
  2. `usesMaterial some Material` (installs piping, thermal insulation, sanitary fixtures).
  3. `worksOn some Project`.

Key Ontological Distinction between Electrician and Plumber: While an `Electrician` has the restriction `worksOn some BuildingProject` (allowing assignment across commercial, industrial, and residential buildings), a `Plumber` has the stricter restriction `worksOn some ResidentialProject`. This distinction is frequently queried during HR compliance audits.

2.3 Directory of Key Personnel and Certifications

Employee ID	Full Name	Ontology Class	Group Classification	Certifications & Assigned Projects
EMP-001	**Ing. Martin Kováč**	`SiteManager`	`ManagementRole`	SKSI License #4412 (Unlimited Height). Manages `EuroveaTower` and `FamilyHouseExample`. Supervises 45 employees.
EMP-002	**Ing. Peter Horváth**	`SiteManager`	`ManagementRole`	SKSI License #5190 (Industrial Structures). Manages Trnava `IndustrialProject` and Bridge `Reconstruction`.
EMP-003	**Ing. arch. Richard Novák, PhD.**	`Architect`	`ManagementRole`	SKA Authorized Architect #1088. Lead on Danube Atrium `OfficeBuilding` (`CommercialProject`); advises client CUST-102.
EMP-004	**Ing. Elena Varga**	`Designer`	Design Staff (`Employee`)	Static & Residential HVAC Specialist. Lead Designer for `FamilyHouseExample` and Tatry `WoodenHouse` (`ResidentialProject`).
EMP-005	**Jozef Marek**	`Electrician`	`ManualWorker`, `ConstructionWorker`	Decree §23 High-Voltage Certification (up to 1000V). Works on `EuroveaTower` and Danube Atrium `BuildingProject`.

EMP-006	**Milan Krátky**	`Plumber`	`ManualWorker`, `ConstructionWorker`	Certified Hydronic Heating & Water Supply Installer. Works exclusively on `ResidentialProject` sites (`FamilyHouseExample`, `EuroveaTower`).
EMP-007	**Štefan Baláž**	`ConstructionWorker`	General Field Crew	Reinforced Concrete Formwork & Masonry Specialist. Assigned to `FamilyHouseExample` (2026) and `EuroveaTower`.

2.4 Mandatory Occupational Health and Safety (HSE) Protocols

All personnel classified under `Employee` must adhere to ApexBuild HSE Directive 2026-04 when entering any active `Project` zone:

1. **Personal Protective Equipment (PPE):** Class EN 397 safety helmets, S3 steel-toe boots, and high-visibility vests are mandatory across all four construction phases (`Planning` site surveys, `Foundation`, `StructuralWork`, and `FinishingWork`).
2. **Site Manager Authority:** No `BuildingProject` may operate without an appointed `SiteManager` (enforced by the ontology axiom `BuildingProject ⊑ hasSiteManager some SiteManager`). If the Site Manager or their certified deputy is absent, crane lifting and deep excavation operations must halt immediately.
3. **Electrical & Plumbing Lockout-Tagout (LOTO):** Every `Electrician` working on a `BuildingProject` must apply padlocked circuit breakers before handling distribution boards. Every `Plumber` working on a `ResidentialProject` must pressure-test manifolds at 1.5x operating pressure (minimum 9.0 bar) prior to concealing pipes under `Insulation` or `Concrete` screed.